

Experiment 5

CHARACTERISTICS OF PASSIVE COMPONENTS

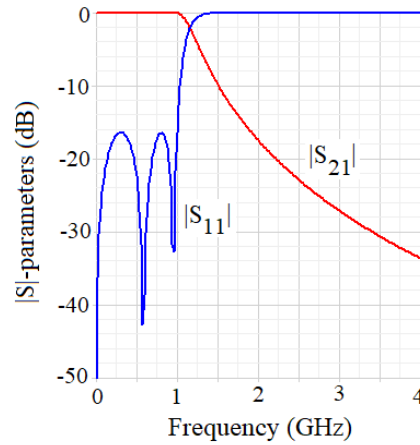
Experiment 5

Objective:

- To calibrate a vector network analyzer (VNA).
- To study the complex scattering (S) parameters of the given passive networks.



(a)

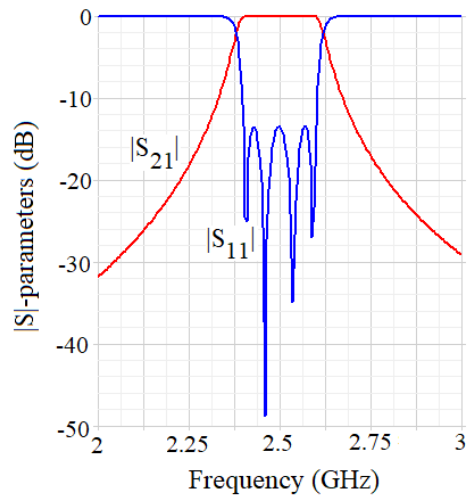


(b)

Fig. 1(a) Layout of a hi-lo lowpass filter and (b) its ideal response.



(a)



(b)

Fig. 2(a) Photograph of a parallel coupled-line band pass filter and (b) its ideal response.

Theory:

The complex scattering parameters of passive and active networks are measured with a vector network analyzer (VNA). A VNA has a two-channel or four-channel microwave receiver, RF source, and internal computing system to process the magnitude and phase of the transmitted and reflected waves from the network under test. The RF

source is usually set to sweep over a specified bandwidth. A four-port reflectometer samples the incident, reflected, and transmitted RF waves. A switch allows the network to be driven from either port 1 or port 2. Four dual-conversion channels convert these signals to IF frequencies (in KHz or MHz), which are then detected and converted to digital form. An internal computer is used to calculate and display the magnitude and phase of the S-parameters or other quantities like SWR, return loss, group delay, impedance, etc. derived from these data. The time-domain response of the network can be obtained by calculating the inverse Fourier transform of the frequency-domain data.

An important feature of the VNA is the substantial improvement in accuracy made possible with error-correcting software. Error is caused by mismatch, imperfect directivity, loss, and variations in the frequency response of the internal components of the VNA as well as external components like cables and connectors. The calibration process determines the error terms, requires a test system consisting of a VNA, cables, and generally a test fixture, and is performed by sequentially making measurements using calibration standards. These calibration standards are one-port and two-port networks such as a 50 Ω matched load, open, short, and through with known characteristics. A full 2-port reflectivity and transmission calibration involves determination of 12 possible systematic error terms. The most common method for correcting for these involves measuring a short, open, and load standards on each of the two ports, as well as transmission between the two ports using a through. The acronym is SOLT. The test port must have the same type of connectors (N-type, 3.5 mm etc.). The SOLT calibration method is suitable for coaxial measurements, where it is possible to obtain the short, open, load and through.

Filters:

A filter is a two-port network used to control the frequency response in an RF or microwave system by providing transmission at frequencies within the passband and attenuation in the stopband of the filter. Typical frequency responses include low-pass,

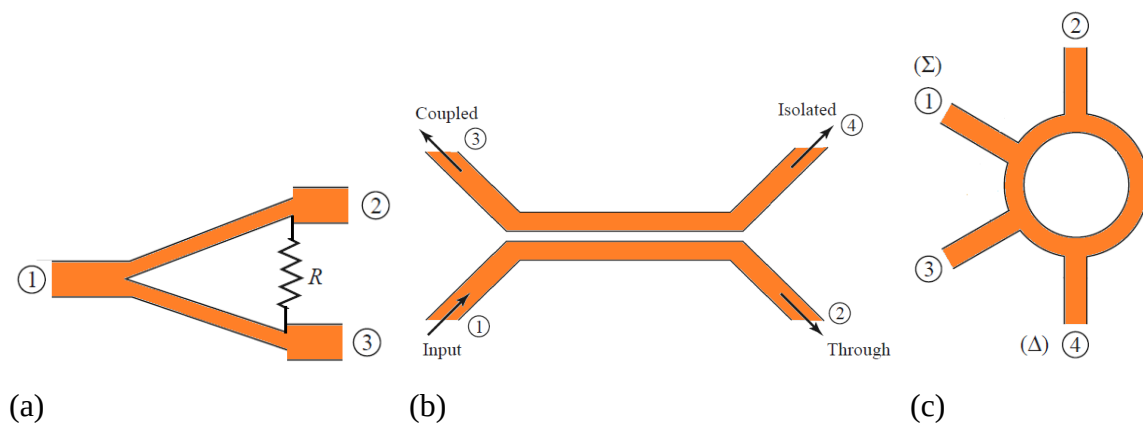


Fig. 3 Microstrip line layout of a (a) Wilkinson power divider, (b) directional coupler, and (c) 180° hybrid coupler.

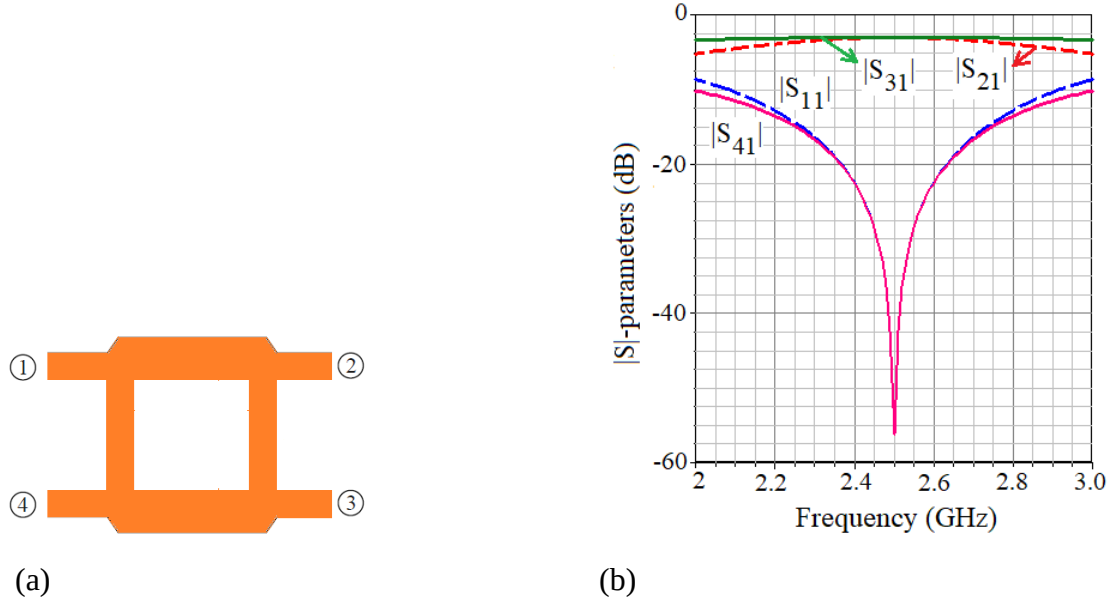


Fig. 4(a) Microstrip line layout of a quadrature hybrid coupler and (b) its ideal response.

high-pass, bandpass, and band-rejection. Filter design usually involves a rigorous network synthesis, called the insertion loss method, which provides a specified frequency response for a given application. Fig. 1 and 2 show typical lowpass and band pass filter in microstrip technology and their responses. Important parameters of a filter are bandwidth of the filter, insertion ($20 \log_{10}|S_{21}|$ dB) and return ($20 \log_{10}|S_{11}|$ dB) losses in the passband, out of band attenuation ($20 \log_{10}|S_{21}|$ dB), and group delay variation ($\tau_d = -\Delta \angle(S_{21}) / \Delta \omega$ ns) within the specified bandwidth. The bandwidth is usually defined by the passband over which ripple value remains constant (e.g. 0.1 dB equal ripple passband).

Power divider and couplers:

Wilkinson power divider:

Among the RF power dividers, 3 dB equal-split Wilkinson power divider is the most popular power divider (Fig. 3a). It can be made with arbitrary power division. The phase difference between the output ports of a power divider is always zero. The S-parameters of a Wilkinson power divider at the design frequency is

$$[S] = \frac{1}{2} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Coupled-line directional coupler:

Coupled-lines directional couplers are a four port device consist of input port, coupled-port, through-port and isolated port. Most of the input power is available at the through port. The phase difference between the through- and coupled-ports at the design

frequency is 90°. They are assumed to operate in the TEM mode, and often implemented in planar technology like stripline, microstrip line, coplanar waveguide, or slotline structures. Coupled transmission lines can support two distinct propagating modes, called the even-mode and odd-mode. The S-parameters of an ideal directional coupler is

$$[S] = \begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix} \quad \text{where } \alpha \text{ and } \beta \text{ are non-zero real numbers.}$$

Rat-race coupler:

The 180° hybrid coupler also known as rat-race hybrid coupler is a four-port network with an 180° phase shift between the two output ports. It can also be operated so that the outputs are in phase. With reference to the 180° hybrid shown in Fig. 3, a signal applied to port 1 will be evenly split into two in-phase components at ports 2 and 3, and port 4 will be isolated. If the input is applied to port 4, it will be equally split into two components with an 180° phase difference at ports 2 and 3, and port 1 will be isolated. When operated as a combiner, with input signals applied at ports 2 and 3, the sum of the inputs will be formed at port 1, while the difference will be formed at port 4. Hence, ports 1 and 4 are referred to as the sum (sigma) and difference (delta) ports, respectively. The scattering matrix for the ideal 3 dB 180° hybrid has the following form:

$$[S] = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

Quadrature hybrid coupler:

Quadrature hybrids are 3 dB directional couplers with a 90° phase difference between the outputs of the through and coupled arms. It is also known as the branch-line coupler. Fig. 4 shows the layout of a quadrature hybrid in microstrip form and its ideal response. The S-parameters of the coupler is

$$[S] = \frac{-1}{\sqrt{2}} \begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix}.$$

Caution:

The VNA is one of the most expensive instruments. Handle with care. Note that all the connectors has a movable part. Only turn this movable parts so that there is no rotational movement of the inner conductors. When connecting the cables it should not provide any pressure on the PCB circuit. Otherwise, it can tear out the SMA connector and the microstrip layout. While handling the microstrip line based circuit, do not touch the circuit layout. Just hold the connectors.

Procedure:

VNA calibration:

1. Set the frequency scale to a suitable range, e.g. 1-3 GHz. Set the display setting to Smith chart.
2. Open calibration option manual, full two port - SOLT.
3. Start calibration using the standard short, open, load and through.
4. After calibration, check the accuracy of calibration by connecting the through. It should show the $|S_{11}|$ as a single point at the center of the Smith chart. Change display setting to log-magnitude. Now the phase of S_{21} should be zero over the range of frequency.

Measurement of the passive devices:

1. After calibrating the VNA, connect the devices one by one.
2. Any unused port should be terminated with a matched load ($50\ \Omega$).
3. Save the s2p measurement data (complex S-parameters) and upload to your email account for future processing.

Calculations:

Filter: Plot the insertion loss and calculate the 3 dB bandwidth of the filter. Plot return loss in the same figure. Plot the time delay τ_d versus frequency and calculate the maximum delay variation within the 3 dB passband.

Couplers: Plot $|S_{11}|$, $|S_{21}|$, $|S_{31}|$, and $|S_{41}|$. Next, plot the amplitude imbalance ($|S_{21}| - |S_{31}|$) and the phase imbalance $\angle S_{21} - \angle S_{31}$. Please note that the phase information is necessary for any coupler. The frequency range for the plots can be set as $f_0 \pm 0.5$ GHz. Calculate bandwidths under the following conditions: 20 dB return loss, 20 dB isolation, ± 0.5 dB amplitude imbalance, $\pm 5^\circ$ phase imbalance. For the rat-race coupler, the measurement has to be repeated for the delta port i.e. plot $|S_{44}|$, $|S_{24}|$, $|S_{34}|$, $|S_{14}|$, $(|S_{24}| - |S_{34}|)$, and $(\angle S_{24} - \angle S_{34})$.

Antenna: Antenna is a one port device. Plot $|S_{11}|$ and find the 10 dB matching bandwidth of the antenna.